
Comparing Schemes and Manifolds

— *EYNTKA* Series —

NATHANAEL CHWOJKO-SRAWLEY

August 22, 2026

Contents

Contents	3
0.0.1 Comparing Schemes and Real/Complex Manifolds	3

This is a quick extract from [1] comparing schemes and smooth manifolds. It is not comprehensive; it is the first things that came to mind. For the full discussion, see the cited notes.

0.0.1 Comparing Schemes and Real/Complex Manifolds

Before proceeding to further explore properties of schemes, it is worth taking a moment to catalogue how schemes and real smooth manifolds have some fundamentally different geometric properties:

1. The connected scheme given by $V(x) \cup V(y, z) \subseteq \mathbb{A}_k^3$ (the plane $x = 0$ together with the x -axis, meeting at the origin) has two irreducible components of different dimensions (as we shall more rigorously show in [1, the section on dimension]). A connected manifold (smooth or even topological) is of a single, well-defined dimension at every point.
2. An abstract manifold cannot self-intersect¹, while schemes (even affine schemes, even varieties) can: consider $V(y^2 - x^3 - x^2) \subseteq \mathbb{A}_k^2$, which has a node at the origin.
3. The patches of a manifold are very regular, being diffeomorphic to $\mathbb{R}^n$; in this sense manifold patches all carry “linear smooth” information, and there is essentially no local diversity from one patch to another. The affine patches of a scheme, by contrast, may be “pathological”. For example, the local rings can be non-reduced, and the coordinate ring of a patch may be non-Noetherian, in which case Spec of it may have uncountably many connected components. (The *local* rings never do: a local ring has only the idempotents 0 and 1, so its spectrum is connected.) This reflects how affine schemes represent “local algebraic information,” which is far richer than the “local smooth linear information” that real manifolds encode. Some of the

¹An *immersion* of a manifold may of course have self-intersections in the target, but the source manifold itself is not singular at such points.

conditions and properties we shall introduce later (e.g. Noetherianity, regularity) will restore some order.

4. In Manifold Theory, the notion of irreducible component is trivial, since in any Hausdorff space the only irreducible subsets are singletons. From an analytic perspective, this existence of non-trivial irreducible components in schemes can be thought of as a consequence of the Zariski topology failing to be Hausdorff, together with the fact that a function vanishing at a non-closed point vanishes on the whole of its closure; part of this phenomenon is the existence of “large” (i.e. non-closed) points for schemes. More generally, the correspondence between irreducible closed sets and their generic points (which holds in every scheme, all of them being sober) shows that schemes encode much more “global” information about codimension subsets than real smooth manifolds do.
5. The topology of $\mathbb{R}^n$ (and hence of $\mathbb{C}^n$ as a real manifold) is the product topology: the sets

$$\prod_{i=1}^n (a_i, b_i), \quad a_i, b_i \in \mathbb{R}, \quad a_i < b_i,$$

form a topological basis for $\mathbb{R}^n$. This is far from the case for $\mathbb{A}_k^n$: the Zariski topology on $\mathbb{A}_k^n$ is *not* the product of the Zariski topologies on $\mathbb{A}_k^1$. The product is strictly coarser: its closed sets are the finite unions of horizontal and vertical lines and points, so a curve such as $V(y - x^2)$, Zariski-closed in $\mathbb{A}_k^2$, is not closed in the product. However, there will still be a meaningful way to construct the scheme $\mathbb{A}_k^n$ from $\mathbb{A}_k^1$ via the *fibered product* (see [1, the section on (fiber) products and fibers]).

6. If a scheme X is irreducible, all its nonempty open sets are dense. Many of the schemes we shall work with will be unions of irreducible subschemes, where for each irreducible subscheme every nonempty affine open subset is dense in that subscheme. Dense open subsets are far from typical for manifolds (e.g. a disjoint union of two manifolds has open subsets contained in a single component, which are not dense).
7. More generally, the behaviour of schemes matches more closely with the behaviour of complex analytic geometry than with smooth real geometry. We shall draw such analogies often: for example, *Hartogs’s Extension Lemma* ([7, chapter 8]) will have parallels with normal schemes, the existence of certain holomorphic/meromorphic functions will tie into divisor theory, compact complex Riemann surfaces will correspond to projective curves, and so forth.
8. (embedding properties) Any smooth real manifold M can be embedded into $\mathbb{R}^N$ for appropriate N (Whitney Embedding Theorem). For classical affine (resp. projective) varieties of dimension n , they essentially by definition embed into $\mathbb{A}_k^N$ (resp. $\mathbb{P}_k^N$) for some N . For complex manifolds, it is *not* the case that they all embed in $\mathbb{C}^{N/2}$, and in particular no compact complex manifold of positive dimension embeds into $\mathbb{C}^N$. This is because complex manifolds possess much more rigidity than real manifolds [3]³. Schemes exhibit this same type of rigidity. The only types of schemes that have a universal embedding theorem into one of the classical spaces are the

²If a compact connected complex manifold does embed holomorphically, it must reduce to a point, since any global holomorphic function on a compact complex manifold is constant. Non-compact complex manifolds that embed *properly* in $\mathbb{C}^N$, equivalently as closed submanifolds, are called *Stein manifolds*, and share many properties of quasi-projective varieties. Properness is essential: $\mathbb{C}^2 \setminus \{0\}$ sits in $\mathbb{C}^2$ as an open submanifold yet is not Stein, by Hartogs.

³For instance, the complex torus $\mathbb{C}/\Lambda$ depends on the lattice Λ up to biholomorphism, giving a one-complex-parameter family of non-isomorphic structures.

complete nonsingular curves, which all embed into a single $\mathbb{P}^3$ however large their genus (see [1, every curve embeds in projective 3-space]). Smooth complete surfaces are still projective, by a theorem of Zariski, and so embed in $\mathbb{P}^5$; it is in dimension three that completeness stops implying projectivity at all: [1, Hironaka’s example] is a smooth complete threefold admitting no embedding into any $\mathbb{P}_k^N$.

By the Kodaira Embedding Theorem [3], a compact Kähler manifold with a positive line bundle can be embedded into projective space. Proper schemes (the algebraic analogue of compact complex manifolds) with an *ample line bundle* (the algebraic analogue of a positive line bundle) similarly embed into projective space. In fact, this means that compact Kähler manifolds with positive line bundle are (as complex analytic spaces) isomorphic to a smooth proper scheme with an ample bundle. This is the closest to a general embedding theorem that is widely used, and will be proven in [1, the section on maps to projective space].

For a closed immersion $X \hookrightarrow \mathbb{A}_R^n$ so that $X \cong V \subseteq \mathbb{A}_R^n$ where V is a closed subscheme, X is topologically a sub-variety, though it may have a more complicated sheaf structure (for more details see [1, the section on abstract varieties]).

For an open embedding $X \hookrightarrow \mathbb{A}_R^n$ so that $X \cong U \subseteq \mathbb{A}_R^n$ where U is open, X is a *quasi-affine subscheme*, which by definition is an open subscheme of affine space. Their characterization is more subtle: the map $X \rightarrow \operatorname{Spec}(\mathcal{O}_X(X))$ must be an open embedding or, equivalently for quasi-compact separated X and as will be discussed in [1, the section on quasi-coherent and coherent sheaves], every quasi-coherent sheaf of $\mathcal{O}_X$ -modules must be generated by global sections. Verifying that every $\mathcal{O}_X$ -module is generated by global sections is difficult; there is a simpler characterization given by the Goodman-Hartshorne Theorem [5], but it is still a nontrivial condition, illustrating the greater complexities that arise when working with non-proper schemes.

More generally, as *Toric Geometry* [4] develops, there are other interesting ambient spaces into which certain types of schemes may be embedded (e.g. toric varieties, Grassmannians, flag varieties, etc.), each serving different purposes depending on the moduli problem under study.

9. Schemes can have “doubled points”: recall that the line with two origins is a standard counterexample for manifolds, as it fails to be Hausdorff at the origin. In scheme theory the analogous phenomenon is permitted; indeed, many moduli spaces that are schemes are naturally non-separated schemes (see [1, the section on Hilbert schemes]).
10. A scheme remembers multiplicity. For example, on a classical point there is little room for interesting functions, but scheme-theoretic functions (i.e. regular functions on $\operatorname{Spec} k[x]/(x^2)$) can encode multiplicity and infinitesimal information. A smooth manifold $(M, \mathcal{A})$ with its maximal atlas does not retain such data.
11. There can be multiple structure sheaves on the same underlying topological space, defining genuinely different schemes (think of the different fields that can serve as the structure sheaf over a single point). In contrast, manifolds of dimension ≤ 3 have unique differential structures up to diffeomorphism (by work of Radó in dimension 2 and Moise in dimension 3); in fact, the categories of piecewise-linear, topological, and smooth manifolds coincide in these dimensions, showing how “linear” manifold theory is in low dimensions. In [1, the section on (fiber) products and fibers] we shall introduce the process of *base change*, by which one can change the structure sheaf without changing the underlying topological space.

12. Though we have not defined an atlas for schemes, [1, the coproduct (pushout) of schemes] will show that schemes can be constructed by gluing affine patches, just as a manifold is built from its atlas, see [1, the section on affine atlases]. ().
13. The natural additional structure on a smooth manifold is a *tangent bundle*, whose transition functions take the form $D(\mathbf{y} \circ \mathbf{x}^{-1})$; that is, they are not just smooth maps between open subsets but must respect a $\mathrm{GL}_n(\mathbb{R})$ -action given by the Jacobian. We shall define the algebraic analogue for schemes (the sheaf of Kähler differentials and the tangent sheaf) in [1, the chapter on sheaves of modules, divisors, and differentials], [1, the section on differentials], though more care is required since schemes are not “locally linear” everywhere in the way smooth manifolds are. Much of the intuition from Poincaré duality (see [6, section 4.3]) will inform the generalization to Serre duality.
14. Schemes can better capture different levels of *local* behaviour. For example, a germ of a smooth manifold at a point is not itself a smooth manifold, but the analogous object for a scheme (the spectrum of a local ring) is again a scheme, called a *local scheme*. This is because commutative rings faithfully encode this local information, and (affine) schemes are the geometric realization of commutative rings.
15. Scheme functions are more “rigid”: on a smooth manifold, partitions of unity exist thanks to the flexibility of the Euclidean topology, but no exact analogue holds for schemes. This rigidity, however, also adds structure and sometimes simplifies definitions (e.g. closed subschemes, introduced in [1, the section on subschemes], are a rather “natural” concept compared to the work that goes into verifying that closed submanifolds are locally diffeomorphic to $\mathbb{R}^k \subseteq \mathbb{R}^n$). As smooth functions on each $U \subseteq M$ form a sheaf, sheaves that admit a “partition of unity” are called *fine sheaves*. A weaker notion of *soft sheaf* will be introduced in [1, the section on sheaf cohomology]; this is the closest scheme-theoretic analogue of the partition-of-unity property.
16. The two theories compute cohomology by different mechanisms, and the previous item is why. On a smooth manifold M the de Rham complex $\Omega_M^\bullet$ resolves the constant sheaf $\mathbb{R}$, and partitions of unity make each Ω_M^p fine, hence acyclic; sheaf cohomology of $\mathbb{R}$ therefore computes de Rham and singular cohomology alike, and the topology of M is recovered from a resolution of a constant sheaf. Neither half of that survives for schemes.

First, there are no fine resolutions available, for want of partitions of unity. Second, the object being resolved is already trivial: on an *irreducible* scheme any two nonempty opens meet, so a locally constant function is constant, every constant sheaf is flasque, and

$$H^i(X, \underline{A}) = 0 \quad \text{for all } i > 0$$

The Zariski topology has too few open sets for a constant sheaf to carry any cohomology at all. Cohomology is therefore rebuilt in two directions, answering two different questions ([1, the chapter on cohomology of schemes]):

- *Coherent cohomology* keeps the topology and changes the sheaf, taking $H^i(X, \mathcal{F})$ for $\mathcal{F}$ quasi-coherent. This measures the scheme structure rather than the underlying space, and is the analogue of Dolbeault cohomology rather than of singular cohomology. Affine schemes are acyclic for quasi-coherent sheaves, so on a separated scheme the Čech complex of any affine cover computes it ([1, the section on Čech cohomology]). The acyclicity is a property of the open sets; on a manifold it runs the other way, coming from softness of the sheaf rather than from the shape of the cover.

- *Étale cohomology* keeps the sheaf and changes the *site*, replacing Zariski opens by étale maps. For n invertible on X this returns the singular answer: over $\mathbb{C}$, $H_{\text{ét}}^i(X, \mathbb{Z}/n) \cong H_{\text{sing}}^i(X(\mathbb{C}), \mathbb{Z}/n)$. See [2].

The finiteness statements match: for X proper over a field and $\mathcal{F}$ coherent, every $H^i(X, \mathcal{F})$ is finite-dimensional, as the Betti numbers of a compact manifold are finite, and both vanish above the dimension.

The two duality theorems are indexed differently. Serre duality ([1, Serre duality for a projective scheme]) pairs H^i with H^{n-i} for $n = \dim X$, where Poincaré duality ([6, section 4.3]) pairs H^i with H^{2n-i} , the real dimension; Hodge theory [3] accounts for the difference by splitting each singular degree into the $H^q(X, \Omega^p)$ with $p + q$ fixed.

Despite the vanishing above, Grothendieck proved that for smooth X over $\mathbb{C}$ the *hypercohomology* of the algebraic de Rham complex $\Omega_{X/\mathbb{C}}^\bullet$, taken in the Zariski topology, computes $H_{\text{sing}}^*(X(\mathbb{C}); \mathbb{C})$. The Poincaré lemma fails Zariski-locally, so $\Omega_{X/\mathbb{C}}^\bullet$ is not a resolution of $\mathbb{C}$ and the statement is about the complex rather than about any one sheaf. [1, the GAGA cohomological dictionary] collects the comparisons.

17. Schemes unify “arithmetic” and “geometry” in a way that manifold theory is not equipped for. As examples, there are no manifold analogues of $\text{Spec } \mathbb{Z}$, $\text{Spec } \mathcal{O}_K$ (where $\mathcal{O}_K$ is a ring of integers of a number field), or $\text{Spec } \mathbb{Z}[t]$. In this way, scheme theory gives geometric meaning to objects that are not tractable in differential geometry. A consequence is that different points of a single scheme can have residue fields of different characteristics (e.g. on $\text{Spec } \mathbb{Z}$, the residue field at the prime (p) is $\mathbb{F}_p$ while the residue field at the generic point is $\mathbb{Q}$), a phenomenon with no manifold analogue. To work in a more “purely geometric” setting, one often restricts to $\bar{k}$ -schemes where $\bar{k}$ denotes the algebraic closure of a field of characteristic 0. In fact, when defining Z -valued points in [1, the section on Z -valued points] for a scheme Z , if $Z = \text{Spec } \bar{k}$ then such points are called *geometric points*. Working over a non-algebraically closed field often signals that the scheme carries arithmetic information.
18. Schemes have non-closed (“generic”) points. Such points are essential: for instance, the ring map $k[x, y] \rightarrow k(x)[y]$ (inverting every nonzero polynomial in x) induces a morphism $\text{Spec } k(x)[y] \rightarrow \text{Spec } k[x, y]$ whose image consists of points lying over the generic point of $\text{Spec } k[x]$. We shall later see that the image of such a morphism is *dense*.
19. Though the above list highlights many differences, many of the usual geometric tools remain applicable in the study of schemes. For example, $\mathbb{C}[x, y]/(x^2 - y^3)$ is not a normal ring. This can be hard to verify algebraically, but viewing $\text{Spec } \mathbb{C}[x, y]/(x^2 - y^3)$ as a curve in $\mathbb{C}^2$ (in the Euclidean topology), the fact that it has a cusp at the origin, and hence is singular there at a codimension-1 point of the curve, is precisely why it fails to be normal. (For a Noetherian ring, normality is Serre’s condition R_1 , regularity in codimension 1, together with S_2 .) This type of geometric reasoning can be powerful: one can study the smoothness (or more specifically the normality) of a variety to determine the integral closure of its coordinate ring.

Though this is a long list consisting mostly of differences, very often in practice (especially in the setting of k -schemes, or when studying curves, surfaces, moduli spaces, etc.) we work with schemes that are proper (the correct generalization of compact manifolds) and have many methods to “resolve” singularities (e.g. eliminate nodes, cusps, etc.) and make the resulting scheme regular⁴. In fact, as we

⁴Though as we shall see, there is no universal resolution of singularities in all characteristics and dimensions.

shall see in [1, the section on GAGA], a result known as *GAGA* (Géométrie Algébrique et Géométrie Analytique) shows that proper schemes over $\mathbb{C}$ with appropriate coherent sheaves are equivalent, as categories, to compact complex analytic spaces with their coherent analytic sheaves. Naturally, arithmetic schemes will not be part of this manifold-scheme correspondence.

Resolution has been completely proved in characteristic 0 (Hironaka) and in low dimensions, but in positive characteristic the general case remains open. Moreover, the singularity information itself is sometimes important and should not always be discarded.

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